

LibraX

Smart Library Management System

Team Information

Team Name	Details
Team	SynCore
Team Head	Amber — https://github.com/Amberparmaar/Project-LibraX-.git
Member 2	Nishaat — https://github.com/nishatafza78600-blip
Member 3	Dua — https://github.com/duaamustafa10
Live Project	https://project-librax.netlify.app

Project Overview

LibraX is a modern Smart Library Management System designed to automate library operations and provide a seamless experience for both administrators and members.

Key Objectives:

- Manage large book collections efficiently
- Automate book issue and return processes
- Provide real-time member information
- Enable QR code-based member verification
- Keep members instantly informed through smart notifications
- Provide instant book information through an AI-powered chatbot

Key Features

Public Portal

- Home, About, Features, Books, and Contact pages

Authentication & User Management

- Secure login and registration
- Firebase-based authentication
- Password reset functionality

Book Management (Admin)

- Add, edit, and delete books
- Inventory and category management
- Advanced search and filter options

Issue & Return Management

- Automated book issuance with due dates
- Quick return processing
- QR code scanning for member verification

Member Features

- View borrowed books
- Digital membership card with QR code
- Profile management and updates

Admin Dashboard

- Real-time analytics and statistics

Notification Feature (New)

- Real-time in-app notifications for members and admins
- Alerts for book due dates and upcoming return deadlines
- Overdue book reminders sent automatically
- Notification when a reserved/requested book becomes available
- Admin alerts for low stock or newly added books
- Confirmation notifications on successful issue and return of a book

Chatbot Feature (New)

- Built-in smart chatbot to assist members and visitors.
- Instantly provides book information on request, including:

Book Title

Author

Price

Available Quantity

ISBN

- Answers member queries about book availability without manual search

Technology Stack

Frontend:

- HTML5, CSS3, JavaScript, Bootstrap 5, Font Awesome, GSAP,

Backend & Database:

- Firebase Authentication, Firestore Database

Libraries & Tools:

- QR code generation & scanning library
- Chatbot logic module for book information queries
- Notification/alert handling module

Database Collections

- Users: Member profiles with authentication and personal information
- Books: Book details including title, author, ISBN, category, price, and quantity
- Issued Books: Records of book issues, returns, and member transactions
- Notifications: Stores due-date alerts, overdue reminders, and system notifications for members and admins
- Contact: Records Contact form info

Pages Overview

- 1. Home Page - Welcome and system overview
- 2. About Page - Library information
- 3. Features Page - System capabilities
- 4. Books Page - Public book catalog

- 5. Contact Page - Communication channel
- 6. Login Page - User authentication
- 7. Register Page - New member registration
- 8. Admin Dashboard - Analytics and controls
- 9. Books Management - Add/edit/delete books (Admin)
- 10. Members Management - Manage member accounts (Admin)
- 11. Issue Book - Process book issuance
- 12. Return Book - Process book returns
- 13. Membership Card - Digital ID with QR
- 14. My Books - View borrowed books (Member)
- 15. Profile - Manage personal information (Member)
- 16. Notifications - View due-date, overdue, and system alerts (New)

Extra Features:

- Chatbot Assistant - Ask for book title, author, price, ISBN and quantity (New)
- QR Scanner - Scan member QR codes
- Generate ID card

Deploy Link

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